

VTPEH 6270 - Check Point 02

Basic R Skills

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1 Introduction

- Background

According to reports in the media, there is a growing proportion of parents of young children, who hesitate or refuse to vaccinate their children in the United States. Several immunizations are state-mandated for children in public schools. Amongst those are Tetanos, Diphteria, Pertussis, Hepatitis B, Measles, Mumps, Rubella, Varicella. Although all states allow children to be exempted from immunization for medical reasons, some states are more stringent than others in allowing exemptions to immunization for philosophical or religious reasons. Under the leadership of the new secretary of health and human services, R. F. Kennedy Jr., federal guidelines for recommended childhood immunizations are shifting and there will be consequences in the stringency of immunization mandates for children attending public schools at the level of states.

- Project Question

Data is available at the CDC that reports on the level of immunization and exemptions granted for medical and non-medical reasons for children enrolled in public Kindergarten classes for the years spanning 2019 to 2024. We will focus first on analyzing the level of non-medical exemptions granted by the various States and follow the evolution of this trend over the years where data is publicly available. We will also look at whether the Covid epidemic had a visible impact on the level of non-medical exemptions granted. Later, we will also inquire about how shifting attitudes towards vaccinations, reflected by shift in states mandates, impact the levels of non-medical exemptions granted. Thirdly, we will analyze outbreaks of diseases that are covered by childhood immunizations and attempt to see if they correlate, as expected, with varied attitudes towards immunizations in the different geographical regions.

- Data Set Overview

We will use the *Vaccination Coverage among Kindergartners* data set available at [https://data.cdc.gov/Vaccinations/Vaccination-Coverage-and-Exemptions-among-Kinderga/ijqb-a7ye/about_data]. The population studied is that of children enrolled in Kindergarten in public schools in the United states. The study reports on vaccination and exemption rates for the school years spanning 2009-2010 to 2024-2025. The data was collected mainly through the census report, which is a mandated collection of data reported by all public school administrators.

2 Loading Data

```
# Set working directory
# C:\Users\alexm\OneDrive\Documents\CLASSES\VTPEH6270\Immunizations_K
setwd("C:/Users/alexm/OneDrive/Documents/CLASSES/VTPEH6270/Immunizations_K")
getwd()

## [1] "C:/Users/alexm/OneDrive/Documents/CLASSES/VTPEH6270/Immunizations_K"

# Load file in df "data_vaccination" from excel file
# downloaded from CDC "Vaccination_Coverage_and_Exemptions_among_Kindergartners_20260201"
data_K <-
  read.csv("data/Vaccination_Coverage_and_Exemptions_among_Kindergartners_20260201.csv",header=TRUE)

# Preview the data
names(data_K)

## [1] "Vaccine.Exemption" "Dose" "Geography.Type"
## [4] "Geography" "School.Year" "Estimate...."
## [7] "Population.Size" "Percent.Surveyed" "Footnotes"
## [10] "Number.of.Exemptions" "Survey.Type"

# count the number of rows
nobs_data_K <- nrow(data_K)
```

The data set is not labeled using best practices (e.g., uses unusual characters like "..."). We clean the formatting using the `janitor` package.

```
# install.packages("janitor")

# library(janitor)
library(janitor)
cleaned_data_K <- clean_names(data_K)

# Preview the cleaned data
names(cleaned_data_K)

## [1] "vaccine_exemption"      "dose"                  "geography_type"
## [4] "geography"              "school_year"           "estimate"
## [7] "population_size"        "percent_surveyed"      "footnotes"
## [10] "number_of_exemptions"   "survey_type"
```

3 Survey the data included in the data set

We count the number of observations in the original data set and also the number of geographical areas surveyed and the type of survey that fed the data set.

```
# count the number of observations
nrow(cleaned_data_K)

# what are the unique values in some of the character columns
unique(cleaned_data_K$geography_type)
unique(cleaned_data_K$survey_type)
```

4 Simplification and survey of data set

*The data set contains data that is either redundant or unnecessary for the analysis planned at this time. We create a new data frame after removing this data that we do not deem necessary at this time. We will go back to the original data set for following analysis.

```
# remove the columns geography_type, footnotes & survey_type

cleaned_data_K <-
  cleaned_data_K[, !names(cleaned_data_K) %in%
    c("geography_type", "footnotes", "survey_type")]

```

Which vaccines are considered in the data set?

```
#the Vaccines/Exemption considered are: unique values in variable vaccine_exemption
unique(cleaned_data_K$vaccine)
```

```
## [1] "Hepatitis B"           "Varicella"             "Polio"                 "Exemption"
## [5] "MMR (PAC)"            "DTP, DTaP, or DT"      "MMR"
```

The considered vaccines or type of exemptions are:

- Hepatitis B

- Varicella
- Polio
- Mumps, Measles, Rubella
- Diphtheria Tetanos Pertussis

When Kindergartners are not vaccinated, an exemption must be submitted:

- Medical Exemption
- Non-Medical Exemption

```
#the dimensions considered are: unique values in variable geography
unique(cleaned_data_K$geography)
```

Which areas are represented in the data set?

The represented geographical areas are:

- Oklahoma, Oregon, Illinois, District of Columbia, Delaware, Kansas, Kentucky, North Carolina, New Mexico, New Jersey, Pennsylvania, Colorado, Montana, New York, Indiana, Louisiana, Utah, Minnesota, Michigan, West Virginia, Washington, Virginia, Missouri, Hawaii, United States, Connecticut, Alaska, Alabama, TX-City of Houston, Mississippi, Nevada, Nebraska, Wyoming, Wisconsin, Idaho, Vermont, Iowa, Georgia, Maine, Massachusetts, Texas, TX-City of San Antonio, Arkansas, Maryland, Ohio, South Dakota, Tennessee, Arizona, North Dakota, New Hampshire, Florida, California, South Carolina, Rhode Island, NY-City of New York, U.S. Median, NY-Rest of state

We will create a new data set that only contains data per state- we will remove the data for the whole USA and the data that is specific to cities (NYC, Houston, San Antonio).

```
# cleaned_data_K %>%
# filter(grepl("^TX-", geography))
class(cleaned_data_K)
```

```
## [1] "data.frame"
```

```
options(max.print = 10000)

library(dplyr)
library(stringr)
cleaned_data_K <- cleaned_data_K %>%
  filter(
    !str_starts(geography, regex("TX-|NY-", ignore_case = TRUE)) &
    !str_detect(geography, regex("United States|U\\.S\\. Median", ignore_case = TRUE))
  )

# count the typical total population surveyed in one school year for one type of Vaccine

library(dplyr)
pop_K <- cleaned_data_K %>%
  filter(vaccine_exemption == "Varicella", school_year == "2024-25",
```

```

      dose == "UTD (unknown disease history)") %>%
    summarise(total = sum(population_size, na.rm = TRUE)) %>%
    pull(total)

# check for no redundancy in the population_size
pop_K_varicella_2024 <- cleaned_data_K %>%
  filter(vaccine_exemption == "Varicella", school_year == "2024-25",
         dose == "UTD (unknown disease history)")

# place commas for the thousands
pop_K = format(pop_K, big.mark = ",")

mean_pop_K <- mean(pop_K_varicella_2024$population_size, na.rm = TRUE)
max_pop_K <- max(pop_K_varicella_2024$population_size, na.rm = TRUE)
mean_pos_K <- format(round(mean_pop_K, 0), big.mark = ",")

```

The data set contains 8166 observations.

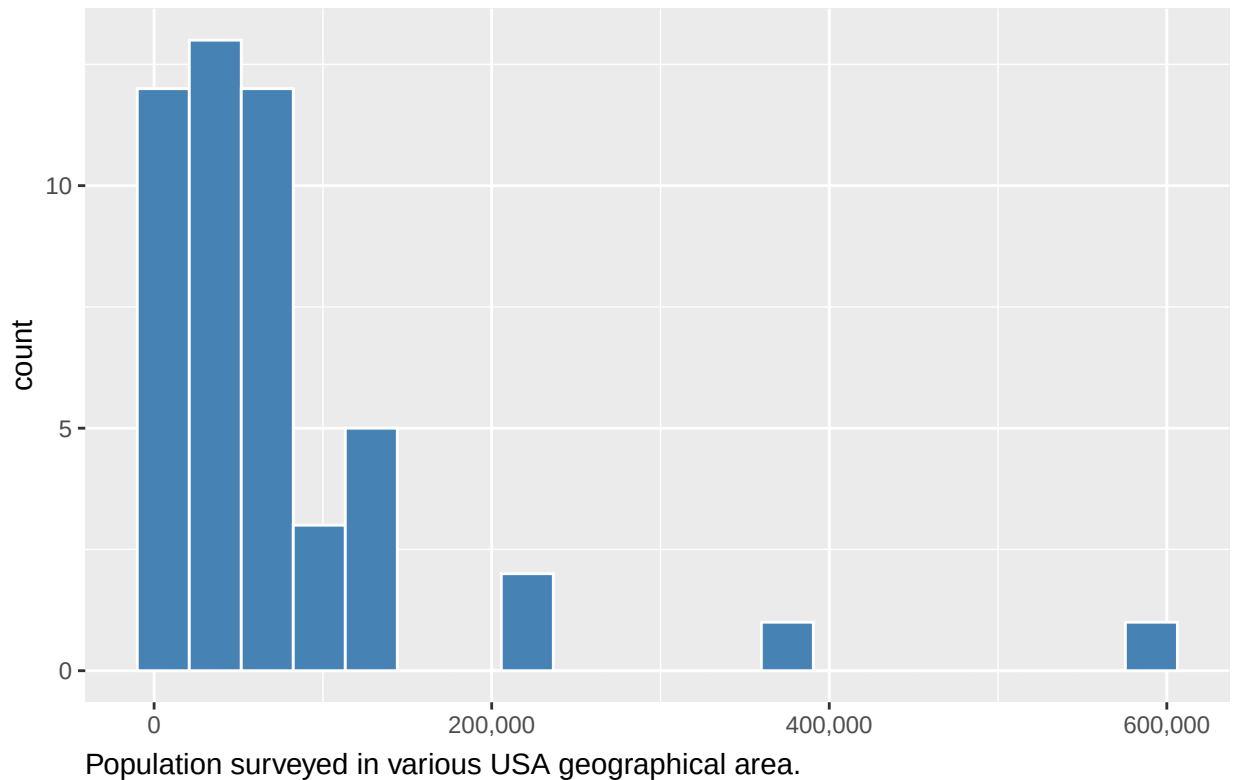
The data set draws from a very large population: for example: the number of children for which Varicella vaccination data was reported in school year 2024-25 is 3,785,644. The average number of children surveyed per geographical area is 7.7258041×10^4 . Let's look at a histogram of population surveyed for Varicella immunization in school year 2024-25.

```

library(ggplot2)

ggplot(pop_K_varicella_2024, aes(x = population_size)) +
  geom_histogram(
    bins = 20,
    fill = "steelblue",
    color = "white",
    na.rm = TRUE
  ) +
  scale_x_continuous(labels = scales::comma) +
  labs(
    x = NULL, # remove x-axis label,
    caption = "Population surveyed in various USA geographical area.
    \n Data is for Varicella 2024-25 school year"
  ) +
  theme(
    plot.caption = element_text(size = 11, hjust = 0)
  )

```



Data is for Varicella 2024–25 school year

What is the time period covered by the study? Indicate all school years sampled

```
# find minimum value of school_year
unique(cleaned_data_K$school_year)

## [1] "2015-16" "2014-15" "2013-14" "2012-13" "2016-17" "2019-20" "2020-21"
## [8] "2009-10" "2021-22" "2010-11" "2011-12" "2018-19" "2017-18" "2023-24"
## [15] "2024-25" "2022-23"

# subtract "_XX" from cleaned_data_K$school_year start at digit 1 end at digit 4
cleaned_data_K$year <- as.numeric(substr(cleaned_data_K$school_year, 1, 4))

# find min and max year of study
min(cleaned_data_K$year)

## [1] 2009

max(cleaned_data_K$year)

## [1] 2024
```

- The study spans from 2009 to 2024. We need to create a new numerical column “year” to describe the 20XX-XX+1 school year as 20XX.

What type of data do you expect the estimated vaccination coverage to be? What is it in the original data set? How can you correct this?

- The estimate value of vaccination coverage is a percentage and thus should be a numerical value. It is encoded in the original data set as a character. It needs to be re-encoded as a numerical value. First step is to remove all the NR or NA values in the estimate column.

```
# Check class
class(cleaned_data_K$estimate)
```

```
## [1] "character"
```

```
# find out data where the data is not numerical
# list rows where estimate is non numerical
# unique(cleaned_data_K$estimate)

# filter out data when non numerical
library(tidyverse)
data_numerical_K <- filter(cleaned_data_K, estimate != "Nreq")
data_numerical_K <- filter(data_numerical_K, estimate != "NReq")
data_numerical_K <- filter(data_numerical_K, estimate != "NR")
# Test: list rows in output data frame where estimate is non numerical
# unique(data_numerical_K$estimate)
```

- The estimate value of vaccination coverage is a percentage and is now ready to be encoded in a numerical value.

```
# Class conversion
# convert estimate column as as numeric
# installing the package dplyr to use the mutate function
library(dplyr)
data_numerical_K <- data_numerical_K %>%
  mutate(estimate = as.numeric(estimate))

# Check output for first line of data after conversion and compare with same before conversion
data1a <- first(data_numerical_K$estimate)
data1b <- first(cleaned_data_K$estimate)

# Check range of values for estimate variable
# find minimum value of estimate value
min(data_numerical_K$estimate)
```

```
## [1] 0
```

```
# find maximum value of estimate value
max(data_numerical_K$estimate)
```

```
## [1] 100
```

- Vaccination coverage is now recorded as a numerical value. The face value of the first line of data was 96.4 before class conversion and is now still 96.4, hence the data was not altered during class conversion.

5 Focus on the Analysis of Vaccination Exemptions

```
# filter the data for exemptions
# using package tidyverse
library(tidyverse)
data_exemption <- filter(data_numerical_K,
                          vaccine_exemption == "Exemption")
data_exemption_nonmed <- filter(data_exemption,
                                dose == "Non-Medical Exemption")

# find minimum value of estimate for Non-Medical Exemptions
min(data_exemption_nonmed$estimate)
# find maximum value of value for exemptions
max(data_exemption$estimate)

# find which geography has minimum value of value for Non-Medical Exemptions
# find year and geography for minimum value of estimate for Non-Medical Exemptions
year_min_exemption <- data_exemption_nonmed[which.min(data_exemption_nonmed$estimate),]
print(year_min_exemption$year)
print(year_min_exemption$geography)

# find which geography has maximum value of value for exemptions
# find year and geography for maximum value of estimate exemption
year_max_exemption <- data_exemption[which.max(data_exemption$estimate),]
print(year_max_exemption$year)
print(year_max_exemption$geography)

# Subset the data to Idaho, that has the highest % of exemptions
data_exemption_Idaho <- filter(data_exemption,
                                geography == "Idaho")
```

6 Table for Vaccination Exemptions

```
# make a nice table for the data
library(kableExtra)
library(gtsummary)
data_exemption %>%
  tbl_summary(
    include = c(vaccine_exemption, dose,
                geography, estimate, population_size, school_year, number_of_exemptions),
    statistic = list(
      all_continuous() ~ "{mean} ({sd})",
      all_categorical() ~ "{n} ({p}%)"
    ),
    missing = "no"
  ) %>%
  modify_caption("Summary of participant characteristics") %>%
  as_kable_extra() %>%
  kable_styling(font_size = 6)
```


Table 1: Summary of participant characteristics

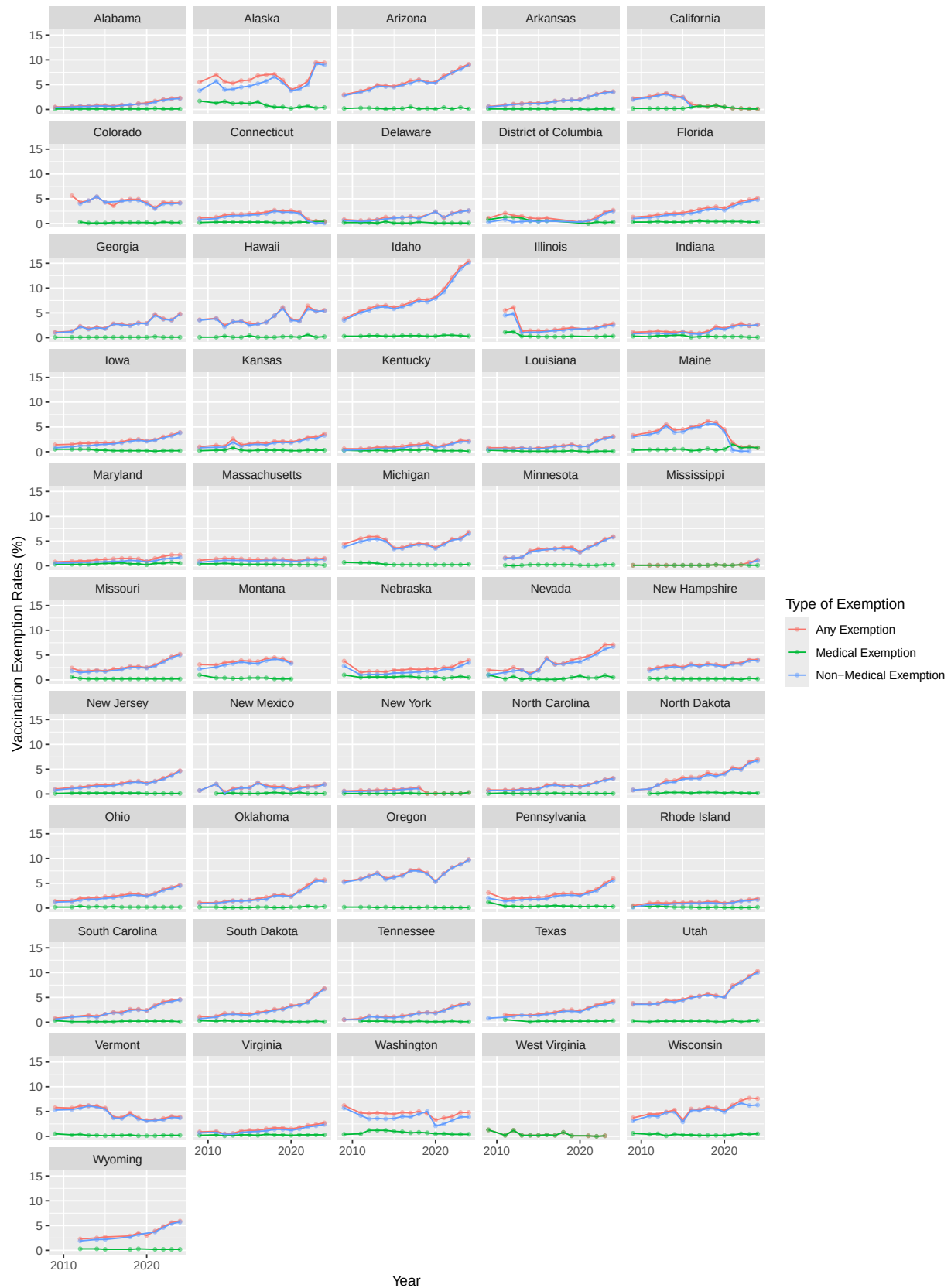
Characteristic	N = 2,167
vaccine_exemption	
Exemption	2,167 (100%)
dose	
Any Exemption	740 (34%)
Medical Exemption	732 (34%)
Non-Medical Exemption	695 (32%)
geography	
Alabama	45 (2.1%)
Alaska	45 (2.1%)
Arizona	45 (2.1%)
Arkansas	44 (2.0%)
California	37 (1.7%)
Colorado	39 (1.8%)
Connecticut	45 (2.1%)
Delaware	42 (1.9%)
District of Columbia	35 (1.6%)
Florida	45 (2.1%)
Georgia	45 (2.1%)
Hawaii	45 (2.1%)
Idaho	45 (2.1%)
Illinois	36 (1.7%)
Indiana	45 (2.1%)
Iowa	45 (2.1%)
Kansas	45 (2.1%)
Kentucky	45 (2.1%)
Louisiana	45 (2.1%)
Maine	44 (2.0%)
Maryland	45 (2.1%)
Massachusetts	45 (2.1%)
Michigan	45 (2.1%)
Minnesota	41 (1.9%)
Mississippi	32 (1.5%)
Missouri	41 (1.9%)
Montana	33 (1.5%)
Nebraska	45 (2.1%)
Nevada	45 (2.1%)
New Hampshire	42 (1.9%)
New Jersey	45 (2.1%)
New Mexico	44 (2.0%)
New York	39 (1.8%)
North Carolina	45 (2.1%)
North Dakota	44 (2.0%)
Ohio	45 (2.1%)
Oklahoma	45 (2.1%)
Oregon	45 (2.1%)
Pennsylvania	45 (2.1%)
Rhode Island	45 (2.1%)
South Carolina	42 (1.9%)
South Dakota	45 (2.1%)
Tennessee	44 (2.0%)
Texas	39 (1.8%)
Utah	45 (2.1%)
Vermont	45 (2.1%)
Virginia	45 (2.1%)
Washington	45 (2.1%)
West Virginia	26 (1.2%)
Wisconsin	45 (2.1%)
Wyoming	28 (1.3%)
estimate	1.96 (2.06)
population_size	77,402 (91,676)
school_year	
2009-10	128 (5.9%)
2011-12	146 (6.7%)
2012-13	146 (6.7%)
2013-14	146 (6.7%)
2014-15	151 (7.0%)
2015-16	151 (7.0%)
2016-17	144 (6.6%)
2017-18	144 (6.6%)
2018-19	146 (6.7%)
2019-20	143 (6.6%)
2020-21	141 (6.5%)
2021-22	144 (6.6%)
2022-23	146 (6.7%)
2023-24	147 (6.8%)
2024-25	144 (6.6%)
number_of_exemptions	1,303 (2,057)

¹ n (%); Mean (SD)

7 Raw plotting of Vaccination Exemptions for each Geographical Area

```
# make a ggplot with multi facets for different states
# plot the data after installing the ggplot2 package
library(ggplot2)
ggplot(data_exemption,
       aes(year, estimate, color=dose)) +
  geom_point(size=1, alpha=0.5) +
  theme(plot.title = element_text(hjust=0.5, size=10)) +
  theme(plot.subtitle = element_text(hjust=0.5, size=10)) +
  labs( title = "Vaccination Exemptions rates for Kindergartners",
        subtitle = "(CDC data)",
        x = "Year",
        y = "Vaccination Exemption Rates (%) ",
        color = "Type of Exemption ") +
  scale_x_continuous(breaks = c(2010, 2020),
                    labels = c("2010", "2020")) +
  facet_wrap(geography ~ ., ncol=5) +
  geom_line()
```

Vaccination Exemptions rates for Kindergartners
(CDC data)



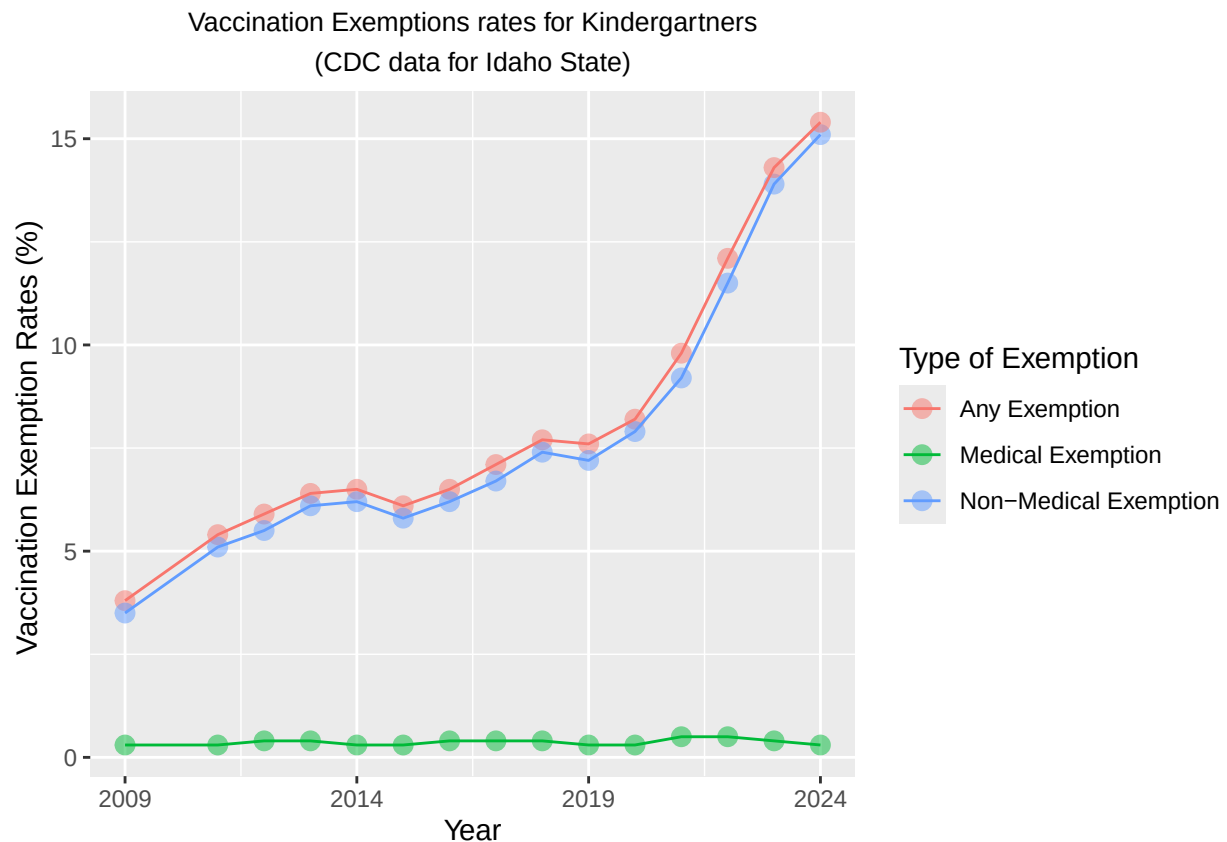
- The geographical area that has the highest level of vaccination exemptions (15.4%) is Idaho in the school year starting 2024. The state that has the lowest level of Non-Medical Exemptions is Maine.

First Subset the data to Idaho, which has the highest % of exemptions

Make a plot showing the estimated vaccination exemption rate across years for Idaho

Generate a **publication-ready** plot.

```
# plot the data after installing the ggplot2 package
library(ggplot2)
ggplot(data_exemption_Idaho,
       aes(year, estimate, color=dose)) +
  geom_point(size=3, alpha=0.5) +
  theme(plot.title = element_text(hjust=0.5, size=10)) +
  theme(plot.subtitle = element_text(hjust=0.5, size=10)) +
  labs(title = "Vaccination Exemptions rates for Kindergartners",
       subtitle = "(CDC data for Idaho State)",
       x = "Year",
       y = "Vaccination Exemption Rates (%) ",
       color = "Type of Exemption") +
  scale_x_continuous(breaks = c(2009, 2014, 2019, 2024),
                    labels = c("2009", "2014", "2019", "2024")) +
  geom_line()
```



```

# find minimum value of estimate for non-medical exemptions for Idaho
data_exemption_Idaho_nonmed <- filter(data_exemption_Idaho,
                                     dose == "Non-Medical Exemption")
print(min(data_exemption_Idaho_nonmed$estimate))
# find maximum value of estimate for non-medical exemptions for Idaho
print(max(data_exemption_Idaho_nonmed$estimate))

# find minimum value of estimate for medical exemptions for Idaho
data_exemption_Idaho_med <- filter(data_exemption_Idaho,
                                   dose == "Medical Exemption")

# write a csv file with only Idaho data

data_Idaho <- subset(data_numerical_K, geography == "Idaho")
write.csv(data_Idaho, "Immunizations_K_Idaho.csv", row.names=F)

```

What does this plot show?

- This plot shows that the vaccination exemption rate in Idaho has steadily increased from a low of 3.5% since data was first collected in 2009, to reach its highest level 15.1% in 2024, which is the last year for which data is available.
- In contrast, the vaccination exemption rate for medical reasons has been relatively stable at rates between 0.3% and 0.5% over the same time period.

For comparison, let's look at the data from Maine, which has the lowest % of Non-Medical Exemptions

```

# Subset the data to Maine, that has the lowest % of Non-Medical Exemptions
data_exemption_Maine <- filter(data_exemption,
                                geography == "Maine")

```

Make a plot showing the estimated vaccination exemption rate across years for Maine

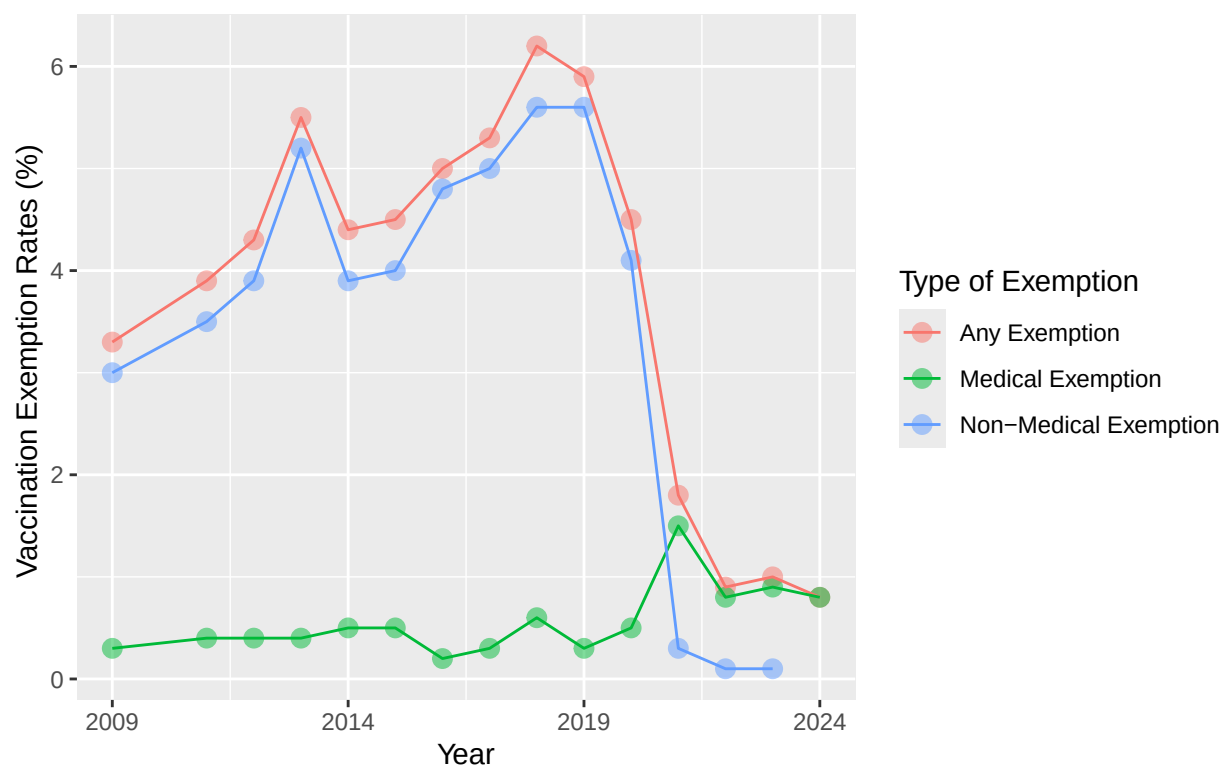
Generate a **publication-ready** plot.

```

# plot the data after installing the ggplot2 package
library(ggplot2)
ggplot(data_exemption_Maine,
       aes(year, estimate, color=dose)) +
  geom_point(size=3, alpha=0.5) +
  theme(plot.title = element_text(hjust=0.5, size=10)) +
  theme(plot.subtitle = element_text(hjust=0.5, size=10)) +
  labs( title = "Vaccination Exemptions rates for Kindergartners",
        subtitle = "(CDC data for Maine State)",
        x = "Year",
        y = "Vaccination Exemption Rates (%) ",
        color = "Type of Exemption ") +
  scale_x_continuous(breaks = c(2009, 2014, 2019, 2024),
                    labels = c("2009", "2014", "2019", "2024")) +
  geom_line()

```

Vaccination Exemptions rates for Kindergartners
(CDC data for Maine State)



```
# find minimum value of estimate for non-medical exemptions for Maine
data_exemption_Maine_nonmed <- filter(data_exemption_Maine,
                                       dose == "Non-Medical Exemption")
print(min(data_exemption_Maine_nonmed$estimate))

# find maximum value of estimate for non-medical exemptions for Maine
print(max(data_exemption_Maine_nonmed$estimate))

# find year for maximum value of estimate for non-medical exemptions for Maine
year_max_exemption_Maine_nonmed <-
  data_exemption_Maine_nonmed[which.max(data_exemption_Maine_nonmed$estimate),]
print(year_max_exemption_Maine_nonmed$year)

# find year for minimum value of estimate for non-medical exemptions for Maine
year_min_exemption_Maine_nonmed <-
  data_exemption_Maine_nonmed[which.min(data_exemption_Maine_nonmed$estimate),]
print(year_min_exemption_Maine_nonmed$year)

# find minimum value of estimate for medical exemptions for Maine
data_exemption_Maine_med <- filter(data_exemption_Maine,
                                   dose == "Medical Exemption")
```

What does this second plot show?

- This plot shows that the vaccination exemption rate in the state of Maine has decreased from a high of 5.6% reached in 2019, to reach its lowest level 0.1% in 2022.

- In contrast, the vaccination exemption rate for medical reasons has been relatively stable at rates between 0.2% and 1.5% over the same time period.

8 AI Use Disclosure Statement

As part of this assignment, please indicate whether you used any AI-based tools (e.g., ChatGPT, Claude, Copilot, Gemini, etc.). Indicate:

- *Did you use AI?*
- I did use ChatGPT.
- *If yes: Write a short disclosure statement (2–3 sentences) describing:*
- I did use ChatGPT to understand error messages and find syntax for given functions.
- *Which tool(s) you used.*
- I used ChatGPT and old Rmarkdown projects to find the correct functions to use and how to use them.
- *How you used it (e.g., to help decide on an analysis approach, to generate a first draft of code, to improve or debug code you had written yourself, etc.).*

Example: “This document was generated using Claude to generate original code, which was then reviewed and adjusted” or “This document was generate using ChatGPT to assist with debugging of code generated by the author”.

Please keep your statement brief and honest. The goal is transparency, not detail: we do not need exact prompts or transcripts, just a clear sense of how AI supported this work.

- This document was generated by resorting to the strategy of trial and error, with the use of google searches and other markdown projects to find available functions and function syntax.